Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/07/2005
PI Name:	Sah, Robert L-Y M.D.		
Project Title:	Fabrication and Growth of Engineered Tissues: Articular Cartilage with Biological and Functional Stratification		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92093	**Congressional District:**	49
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
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No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhatia, Sangeeta (University of California, San Diego) Hoger, Anne (University of California, San Diego) Klisch, Stephen (California State University, San Luis Obispo) Masuda, Koichi (Rush Medical College) Price, Jeff (University of California, San Diego) Schumacher, Barbara (Rush Medical College) Thonar, Eugene (Rush Medical College)		
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Task Description:	A fundamental basis of tissue engineering is the ability to direct heterogeneous cell populations to form specialized tissues. The overall hypothesis of the proposed research is that controlled patterning of distinct cell populations allows the fabrication and growth of tissue with multicellular units of desired mechanical function. The general objectives are (1) to develop advanced technologies for targeting cells and thereby enable the fabrication of stratified tissues with increasing precision and (2) to establish and extend quantitative theory and experimentation that provides a foundation for engineering the growth of mechanically functional tissues through biophysical regulation. (1) Pattern distinct chondrocyte populations in order to form cartilage constructs with multi-cellular architectures of increasing precision. (2) Develop a quantitative mixture model of tissue growth as well as relationships among tissue biomechanical properties, biochemical composition, and structure. Limitations of current tissue engineering strategies for articular cartilage repair are (1) failure to recapitulate the normal stratified structure of articular cartilage and (2) the lack of a model of cartilage growth that accounts for biomechanical evolution of tissue properties in relationship to tissue composition and structure. The ability to form more precise cartilage constructs and the ability to describe the detailed growth of cartilage will form fundamental technologies and theories that are likely to help increase the success of engineering functional and lasting cartilage tissue.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	An review chapter of cartilage repair, regeneration, and replacement, has been published (Sah et al, 2005). Using sedimentation, stratified tissue constructs from selected populations of chondrocytes have been formed. The depth-varying mechanical properties of stratified and mixed cartilage constructs have been evaluated. Such constructs exhibit a variation in compressive modulus, increasing with depth from free surfaces, a pattern characteristic of normal articular cartilage (Klein et al, 2004). Using BioMEMS (microelectromechanical systems), methods have been developed to pattern selected chondrocyte populations. A theoretical framework has been advanced to define how dielectrophoretic methods can be used to target particles with distinct dielectric properties (Albrecht et al., 2004). The dielectrophoretic method has been demonstrated to be applicable to cells (Albrecht et al., 2005), including chondrocytes (Albrecht et al., 2005). A variety of techniques have been advanced for assessing the effects of cell organization on growth of cartilaginous tissue. A method for analyzing the 3-dimensional organization of cells in cartilage has been developed (Jadin et al., 2005). The method has been extended to analyzing the interface between cartilage and bone (Lewis et al., 2005). A method for labeling and tracking chondrocytes has been developed (Chawla et al., 2004). When remodeling results in a mechanically deficient interface, local mechanical properties are disrupted (Bae et al., 2004). Interface and local tissue properties can also be analyzed by a new tissue preparation and testing method (Gratz et al., 2005). A growth mixture model has been developed to describe and analyze a variety of situations. For infinitesimal strains, solutions to selected boundary-value problems have been determined (Klisch et al., 2005). Also, for finite strain conditions, a theoretical formulation has been developed and solved for certain geometries and boundary values (Klisch et al., 2005). Such a model formally describes the relationship between matrix metabolism, mechanical regulation, and evolution of mechanical properties, and was applied to measures of the mechanical properties of bovine articular cartilage during growth.
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